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G H Raisoni International Skill Tech University, Pune

F.Y. B. Tech (Sem-II)

CAE II- Summer 2026 (2025 Pattern)

DIGITAL LOGIC AND DESIGN (25UESL1105)

(Time: - 01 Hour) **(Max. Marks: 20)**

Instructions to the students:

i) Solve Q1 or Q2 and Q3 or Q 4
ii) Neat diagrams must be drawn wherever necessary.
iii) Figures to the right indicate full marks.

Q.No		*Question*	Marks	CO	BL
Q1.	(a)	Explain sequential circuits in digital electronics? Explain their working principles and classify the different types of sequential circuits with brief descriptions.	[5]	2	2
	(b)	Compare the working of SR flip-flop and JK flip-flop. Explain their operation with the help of block diagram, truth table, circuit diagram, and typical applications.	[5]	2	3
		OR			
Q 2.	(a)	Explain shift registers in sequential circuits? Explain their working principle and discuss the different types of shift registers with brief descriptions.	[5]	2	2
	(b)	Design a 3-bit synchronous up counter and explain its operation using the state table (truth table), logic circuit diagram, and timing diagram.	[5]	2	3
Q 3.	(a)	Explain the concept of an Algorithmic State Machine (ASM). Describe the basic components of an ASM chart in brief.	[5]	3	2
	(b)	Explain the working of Mealy and Moore machines. Highlight at least three differences between them in terms of their features and applications.	[5]	3	2
		OR			
Q4.	(a)	Explain the state reduction technique in an Algorithmic State Machine (ASM). Illustrate the method with a suitable example.	[5]	3	2
	(b)	Highlight any five applications of Finite State Machines (FSM) with brief explanations.	[5]	3	2